

Syllabus and Schedule of ECE 2312: Discrete-Time Signal and System Analysis

Course Schedule and Reading Assignments

Date	Day	Topic	Notes	HW/Project
01/10	Tuesday	Course Intro/Overview Discrete Time Signals & Systems — What and Why?	pre-note post-note	
01/11	Wednesday	Classes of Signal and Signal Operations	pre-note post-note	
01/12	Thursday	Linear Time-Invariant Systems	pre-note post-note	
01/13	Friday	Convolution Sum	pre-note post-note	
01/16	Monday	NO CLASS (Martin Luther King Day)		
01/17	Tuesday	Coding Session 1	codes	Assignment 1 Online
01/18	Wednesday	Graphical Convolution Sum	pre-note post-note	
01/19	Thursday	Quiz 1		
01/20	Friday	Convolution and LTI System Properties	pre-note post-note	Project 1 Online
01/23	Monday	Problem Solving Class		
01/24	Tuesday	Discrete Fourier Series	pre-note post-note	
01/25	Wednesday	Bilateral Z-Transform	pre-note post-note	
01/26	Thursday	Inverse Z- Transform	pre-note post-note	Quiz 2 Online
01/27	Friday	One-Sided Z-Transform	pre-note post-note	Assignment 1 Due
01/30	Monday	Discrete Time Fourier Transform (DTFT)	pre-note post-note	Assignment 2 Online

Date	Day	Topic	Notes	HW/Project
01/31	Tuesday	Discrete Time Fourier Transform Properties	pre-note post-note	
02/01	Wednesday	NO CLASS (Wellness Day)		
02/02	Thursday	Inverse Discrete Time Fourier Transform (IDTFT) LTI System Frequency Response	pre-note post-note	Project 1 Due
02/03	Friday	Problem Solving Class		Project 2 Online
02/06	Monday	Coding Session 2	pre-note post-note	Quiz 3 Online
02/07	Tuesday	Discrete Fourier Transform & Properties	pre-note post-note	
02/08	Wednesday	Inverse Discrete Fourier Transform	pre-note post-note	
02/09	Thursday	Fast Fourier Transform	pre-note post-note	
02/10	Friday	Discrete-Time System Implementation & Quantization	pre-note post-note	Assignment 2 Due
02/13	Monday	Sampling of Continuous-Time Systems	pre-note post-note	Assignment 3 Online
02/14	Tuesday	Reconstruction Using Interpolation	pre-note post-note	
02/15	Wednesday	Aliasing	pre-note post-note	
02/16	Thursday	Problem Solving Class	pre-note post-note	Project 2 Due
02/17	Friday	Ideal Filters		Project 3 Online
02/20	Monday	IIR Filter Design	pre-note post-note	Quiz 4 Online
02/21	Tuesday	System Identification: FIR Model	pre-note post-note	
02/22	Wednesday	Coding Session 3	pre-note post-note	
02/23	Thursday	NO CLASS (Academic Advising Day)		
02/24	Friday	NO CLASS (Wellness Day)		
02/27	Monday	FIR Filter Design	pre-note post-note	Assignment 3 Due
02/28	Tuesday	Problem Solving Class		

Date	Day	Topic	Notes	HW/Project
03/01	Wednesday	Zero-Phase Digital Filters	pre-note post-note	
03/02	Thursday	Coding Session 4	pre-note post-note	Quiz 5 Online Project 3 Due
03/03	Friday	Wavelets and Filter Banks	pre-note post-note	

Detailed Course Content Matrix

	Monday	Tuesday	Wednesday	Thursday	Friday
Week 1		January 10 DTST — What and Why?	January 11 Signal Class	January 12 LTIS	January 13 Convolution Sum
Week 2	January 16 NO CLASS	January 17 Coding Session 1	January 18 Graphical Convolution Sum	January 19 Quiz 1	January 20 Conv & LTI Properties
Week 3	January 23 Problem Solving	January 24 DFS	January 25 Z-Transform	January 26 Inverse Z-Transform Quiz 2	January 27 One-Sided Z-Transform Assignment 1
Week 4	January 30 DTFT	January 31 DTFT Properties	February 1 NO CLASS	February 2 IDTFT LTI Frequency Response Project 1 Due	February 3 Problem Solving
Week 5	February 6 Coding Session 2 Quiz 3	February 7 DFT Properties	February 8 IDFT	February 9 FFT	February 10 DTS Quantization Assignment 2
Week 6	February 13 Sampling	February 14 Interpolation	February 15 Aliasing	February 16 Problem Solving Project 2 Due	February 17 Ideal Filter
Week 7	February 20 IIR Filter Quiz 4	February 21 FIR Model	February 22 Coding Session 3	February 23 NO CLASS	February 24 NO CLASS
Week 8	February 27 FIR Filter Design Assignment 3	February 28 Problem Solving	March 1 Zero-Phase Digital Filters	March 2 Coding Session 4 Quiz 5 Project 3 Due	March 3 Wavelets and Filter Banks CLASS ENDS